UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CHEMISTRY **5070/03**

Paper 3 Practical Test
CONFIDENTIAL INSTRUCTIONS May/June 2004

Great care should be taken that any confidential information given does not reach the candidates either directly or indirectly.

INSTRUCTIONS TO SUPERVISORS

Teachers responsible for the examination are **NOT** allowed to consult the question paper before the day of the examination. However, they are asked to carry out a titration between solutions **P** and **Q**, to ensure that the concentrations of the two solutions fall within the given range.

On the day of the examination, the Supervisor is asked to perform the experiments in **Questions 1 and 2** and to record the results on a spare copy of the question paper clearly labelled "Supervisor's Results", followed by the number of the Centre. This must be enclosed with the scripts. **Unless this is done candidates may be unavoidably penalised**.

It is essential that candidates accept the descriptions of the solutions as they appear in the question paper.

If candidates from more than one Centre are taking the examination, it is essential that a copy of the Supervisor's Results should be sent with the scripts for each Centre.

If you have any problems or queries regarding these Instructions, please contact CIE
by e-mail: International@ucles.org.uk,
by phone: +44 1223 553554,
by fax: +44 1223 553558,
stating the nature of the query and the syllabus number quoted above.

This document consists of **5** printed pages and **3** blank pages.

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INSTRUCTIONS TO SUPERVISORS

For Question 1

Candidates will require the following.

(a) Approximately 2.0 mol/dm^3 hydrochloric acid (HC*l*), labelled **P**.

Allow each candidate approximately 200 cm^3.

(b) Approximately 2.0 mol/dm^3 sodium hydroxide (80 g/dm^3 NaOH), labelled **Q**.

Allow each candidate approximately 200 cm^3.

Note: Some variation in the above concentrations is acceptable but it is **essential** that 25.0 cm^3 of **Q** reacts with between 24.0 cm^3 and 26.0 cm^3 of **P** when titrated with an indicator in the normal manner.

P and **Q** should each be at laboratory temperature at the start of the examination.

It is essential that candidates accept the descriptions of these solutions on the question paper.

The following apparatus should be provided for each candidate;
a 50 cm^3 burette;
a 50 cm^3 or 100 cm^3 measuring cylinder;
an expanded polystyrene (or other light plastic) cup of at least 150 cm^3 capacity supported in a suitable beaker;
a 0 °C to 100 °C, or 0 °C to 50 °C, thermometer graduated at 1 °C.

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INSTRUCTIONS TO SUPERVISORS

For Question 2

Each candidate will require the following.

(a) A solution prepared by dissolving 50 g of hydrated copper(II) sulphate, $CuSO_4.5H_2O$, in 1 dm^3 of distilled water and then adding dilute aqueous ammonia until the precipitate just dissolves (approximately 400 cm^3 of 2.0 mol/dm^3 aqueous ammonia). The final solution should be labelled **S**. If the final solution appears cloudy, it should be filtered before use.

Allow each candidate approximately 30 cm^3.

(b) Access to

(i) approximately 1.0 mol/dm^3 hydrochloric acid,

(ii) approximately 1.0 mol/dm^3 nitric acid,

(iii) approximately 0.2 mol/dm^3 aqueous barium nitrate (or approximately 0.2 mol/dm^3 aqueous barium chloride, labelled 'barium nitrate'),

(iv) approximately 0.05 mol/dm^3 aqueous silver nitrate,

(v) approximately 0.1 mol/dm^3 aqueous potassium iodide,

(vi) approximately 0.2 mol/dm^3 aqueous sodium thiosulphate,

(vi) the usual reagents needed to test for the gases mentioned in the syllabus, including limewater, approximately 0.1 mol/dm^3 aqueous potassium dichromate(VI), $K_2Cr_2O_7$, red and blue litmus paper or Universal Indicator paper, and splints.

(c) A supply of test-tubes, approximately 125 mm $\times$ 16 mm.

(d) A boiling tube, approximately 150 mm $\times$ 24 mm.

(e) A stirring rod.

It is advisable to issue candidates with safety goggles.

In both questions, more material may be issued without penalty, if required, but this should not be necessary.

The standard Report Form to be included with the scripts is given on pages 7 and 8. Please detach and enclose it with the scripts in the normal way.

See also the side lined notes on pages 1, 7 and 8.

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This form must be completed and returned in the envelope with the scripts.

REPORT ON PRACTICAL CHEMISTRY

ORDINARY LEVEL, JUNE 2004

1 (a) *Supervisor's results*

Supervisors must use a spare copy of the question paper to report their results for **Q.1 and Q.2** and enclose this copy of the question paper with the candidates' answers. This copy of the question paper should be clearly labelled 'Supervisor's Results'. Failure to enclose these results and this report form may lead to candidates being unavoidably penalised.

If candidates from more than one Centre are taking the examination, it is **essential** that a copy of the 'Supervisor's Results' should be sent with the scripts from **each Centre**. At larger centres where scripts are to be despatched in more than one envelope, it is essential that a copy of the Supervisor's Results is enclosed in each envelope.

(b) The index number of the candidates in each session were as follows.

First session	Second session

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2 The Supervisor is invited to report details of any difficulties experienced by candidates, giving names and index numbers.

This report should include reference to:

(a) any general difficulties encountered in making preparation;

(b) difficulties due to faulty apparatus or material;

(c) accidents to apparatus or materials.

Other cases of individual hardship, e.g. illness, temporary disability, should be reported direct to UCLES on the normal 'Application for Special Consideration' form.

NAME OF CENTRE .....

SIGNED .....
Supervisor

CENTRE NUMBER .....

If the candidates' Centre number is different from the number of the Centre at which the examination was taken, the Supervisor should write **both Centre numbers in the space provided.**

Declaration (to be signed by the Principal).

The preparation of this Practical examination has been carried out so as to maintain fully the security of the examination.

SIGNED .....

NAME (in block capitals) .....

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